

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) Finds mean	AO1.1b	B1	Mean $\bar{X} = \frac{3046.14}{100} = 30.46(14)$
	Finds standard deviation awfw (4.17 – 4.20) Allow $s = \sqrt{\frac{1746.29}{100}} = 4.179$	AO1.1b	B1	Standard deviation $s = \sqrt{\frac{1746.29}{99}} = 4.20$
	(ii) States max range for a normal distribution	AO1.2	M1	$\bar{X} + 3s \approx 43$ $\bar{X} - 3s \approx 18$
	Explains why normal model acceptable	AO2.4	E1	Normal dist model acceptable because data can be regarded as continuous and the range of amounts is within $\bar{X} \pm 3s$
(iii)	Finds probability awfw (0.095 – 0.097)	AO1.1b	B1	$P(X < 25) = 0.0967$
(b)	Finds z from calculator	AO1.1b	B1	$P\left(Z < \frac{30 - 29.55}{\sigma}\right) = 0.55$
	Uses Normal model to form equation	AO3.4	M1	$\frac{30 - 29.55}{\sigma} = 0.1257$
	Finds correct sd Correct accuracy required for this mark Accept 3.5 or 3.7	AO1.1b	A1	$\sigma = \frac{30 - 29.55}{0.1257} = 3.6$
				Total 8 marks

Q2.

Length $L \sim N(69.5, 0.55^2)$

(a) (i) $P(L < 70) = P\left(Z < \frac{70 - 69.5}{0.55}\right) =$

Standardising (69.5, 70 or 70.5) with 69.5 and ($\sqrt{0.55}$, 0.55 or 0.55^2) and/or $(69.5 - x)$

M1

$$P(Z < 0.91) =$$

0.91 AWRT; ignore sign

A1

$$0.818 \text{ to } 0.82(0)$$

AWFW (0.81835)

A1

3

(ii) $P(69 < L < 70) =$

$$P(L < 70) - P(L < 69) =$$

$$P(Z < 0.91) - P(Z < -0.91) =$$

Difference (70 – 69)

M1

$$P(Z < 0.91) - \{1 - P(Z < 0.91)\} =$$

$$(0.81835) - (1 - 0.81835) =$$

Correct area change

m1

$$0.636 \text{ to } 0.64(0)$$

AWFW (0.63670)

A1

3

(iii) $P(L = 70) = 0$

CAO

B1

1

(b) $0.90 \text{ (90\%)} \Rightarrow z = -1.28$

AWRT; ignore sign (-1.2816)

B1

$$z = \frac{l - 69.5}{0.55}$$

Standardising l with 69.5 and 0.55;
allow (69.5 – l)

M1

$$= \pm 1.28(16)$$

Equating z-term to the z-value

A1

Hence $\bar{l} = 68.7$ to 68.9

AWFW; CSO (68.796)

A1

4

(c) Variance of $\bar{L}_{20} = \frac{0.55^2}{20} = 0.0151(25)$

SD of $\bar{L}_{20} = \frac{0.55}{\sqrt{20}} = 0.123$

CAO/AWRT; stated or used

B1

$$P(\bar{L}_{20} > 69.25) = P\left(Z > \frac{69.25 - 69.5}{\sqrt{0.55^2/20}}\right)$$

*Standardising 69.25 with 69.5 and 0.123;
allow (69.5 – 69.25)*

M1

$= P(Z > -2.03) = P(Z < 2.03) =$
Correct area change

m1

0.978 to 0.98(0)

AWFW (0.97896)

A1

4

[15]

Q3.

(a) (i) $X \sim N(\mu, 4^2)$

$\mu = 106$

$P(X < 110) = P\left(Z < \frac{110 - 106}{4}\right)$

Standardising (109.5, 110 or 110.5) with

106 and ($\sqrt{4}$, 4 or 4^2) and/or ($106 - x$)

M1

$$= P(Z < 1)$$

CAO; ignore sign

A1

$$= 0.841$$

AWRT (0.84134)

A1

3

(ii) $P(\text{underweight}) = P(X < 100)$

Use of AFWW 99 to 100

M1

$$= P(Z < -1.5) = 1 - \Phi(1.5)$$

Area change

m1

$$= 1 - 0.93319 = 0.0668 \text{ to } 0.067$$

AWFW (0.06681)

A1

3

(b) $2\% \Rightarrow z = -2.0537$

AWFW 2.05 to 2.06; ignore sign

B1

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$$z = \frac{100 - \mu}{4}$$

M1

Thus $\frac{100 - \mu}{4} = -2.0537$

m1

Thus $\mu = 108.2 \text{ to } 108.3$

A1

4

[10]